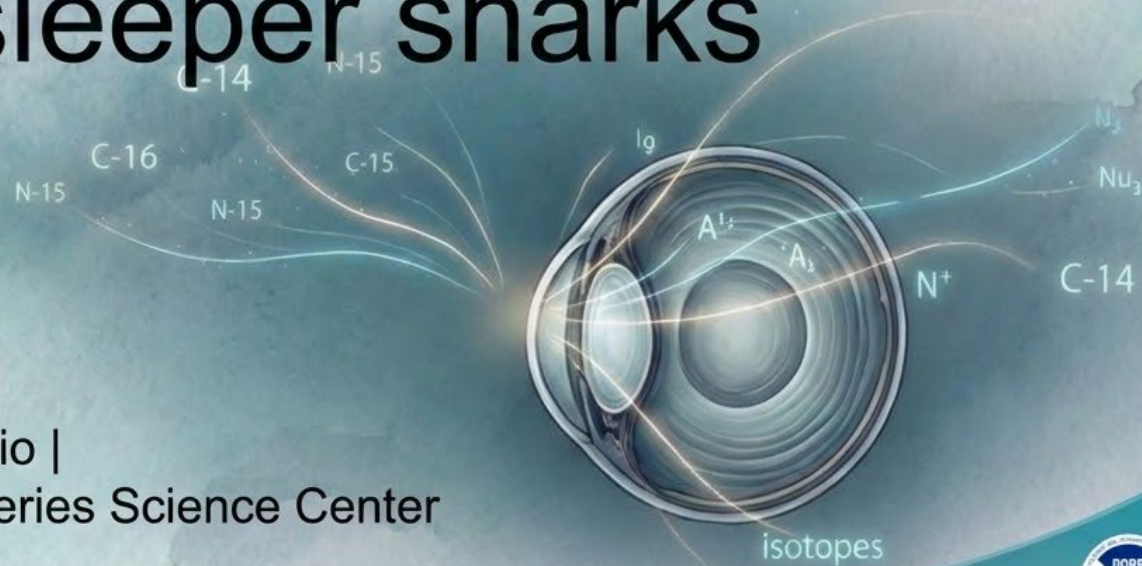


Utility of eye lens chemistry to inform the biology of Pacific sleeper sharks



Old Sharks



Cindy Tribuzio |
Alaska Fisheries Science Center



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NPRB 2301: Ageing of Pacific Sleeper Shark



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ALASKA BIOMAP
scientific services



Outline

NPRB 2301 Objectives

Estimate plausible ranges of age-at-length, age-at-maturity, and lifespan of PSS from 14C.

Investigate sources of potential variation in isotopic values of shark eye lens protein.

Estimate age-related life history parameters and test more robust stock assessment approaches.

Outline

Species of interest

Eye lens chemistry 101

14C background

Preliminary results

Next steps

NPRB 2301 Objectives

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Species of interest



Photo Credit: IPHC

Species of interest

Pacific sleeper shark

Incidentally caught in fisheries

Not targeted

Extremely data limited

Photo Credit: IPHC

Species of interest

Pacific sleeper shark

Incidentally caught in fisheries

Not targeted

Extremely data limited

They get REALLY big

For a great overview of the species:

[Matta et al. 2023 Western Groundfish Conference.](#)

[Matta et al. 2024. Polar Biology article](#)



Picture: Clark et al. 1990

Sugura Bay, Japan

Est size: 660-720 cm TL Matta
et al. 2024

Species of interest



Pacific spiny dogfish

Most common shark in eastern North Pacific

Small shark with large shark life history

Subject of many ageing studies

Used as a proxy species in this study with some bonus analytics

Photo credit: National Ocean Service



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What is eye lens chemistry?

Eyes are like onions, they have layers

Layers sequester chemicals as they form

Provides environmental or biological information at the time of layer formation

Core represents early development, outermost layers is more recent

Can be used to see trends over the lifetime of the animal



What is eye lens chemistry?

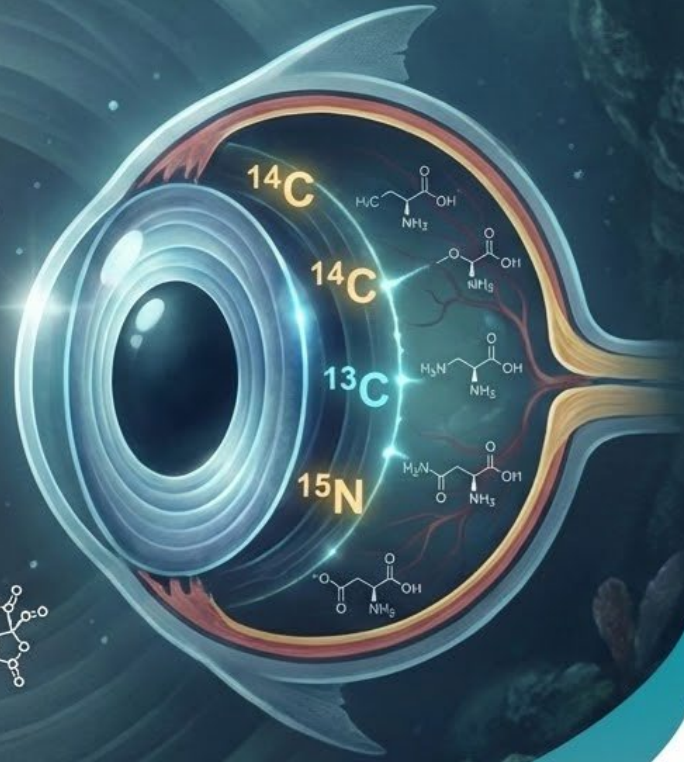
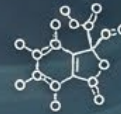
For the purposes of this study

^{14}C : bomb produced radiocarbon in the environment as a chronometer

^{13}C : stable isotope signal of habitat or region

^{15}N : stable isotope signal of trophic position

CSIA-AA: compound specific isotopic analysis of amino acids, high resolution ecology



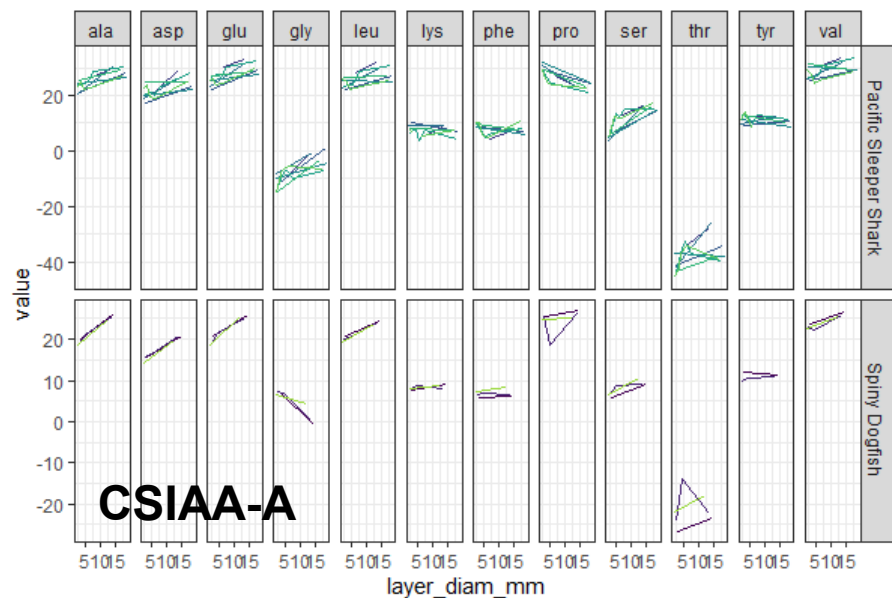
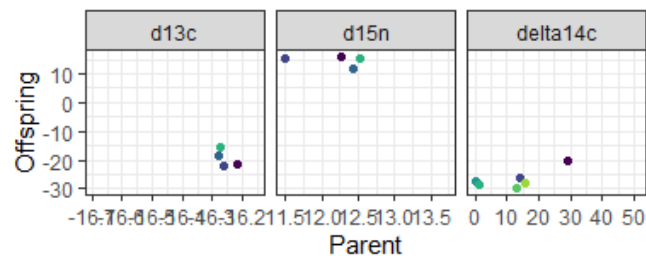


That's a lot to unpack

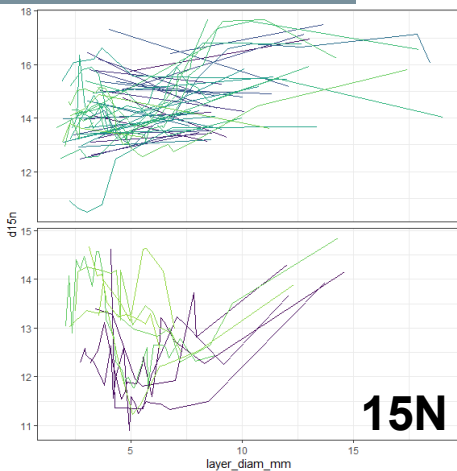
65 Pacific Sleeper Shark, 451 layers

12 Spiny Dogfish, 216 layers

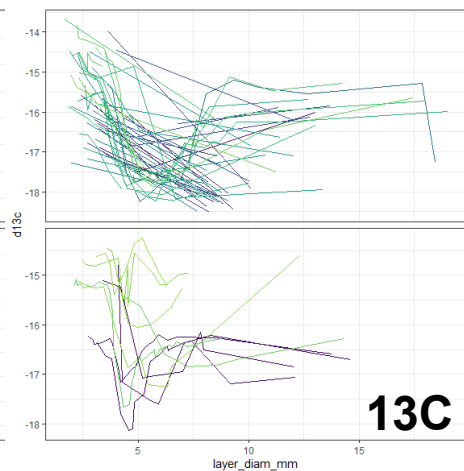
Maternal



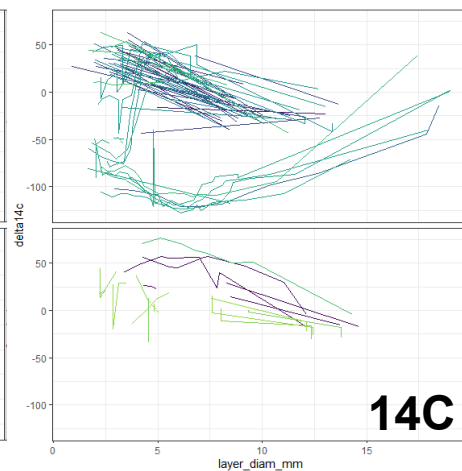
CSIAA-A



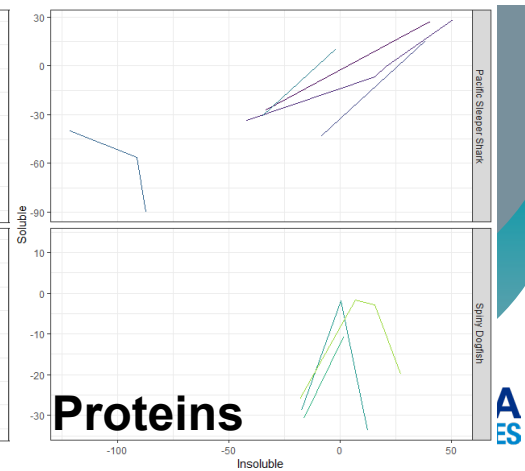
15N



13C



14C



Proteins

Pacific Sleeper Shark

Spiny Dogfish

AES

What is eye lens chemistry?

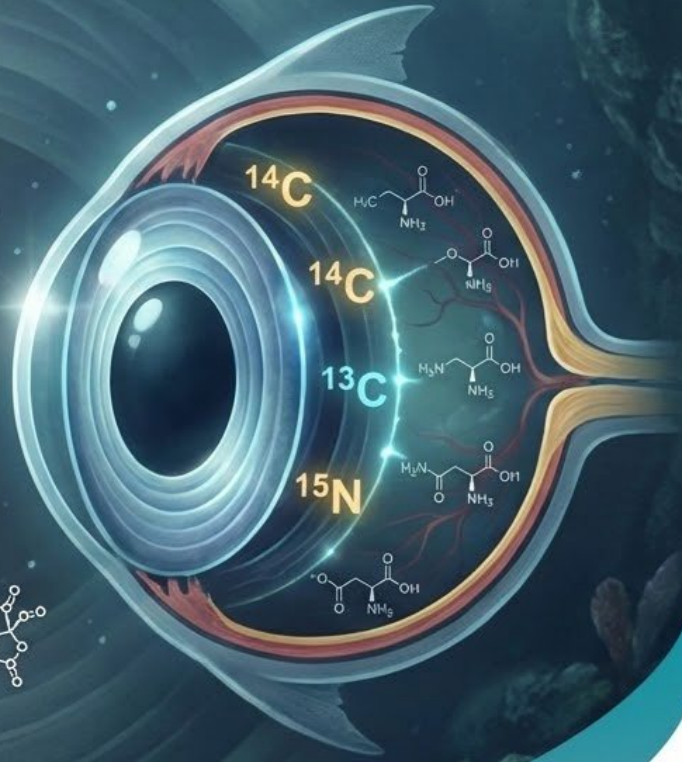
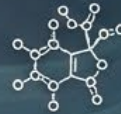
For the purposes of this study ~~TALK~~

14C: bomb produced radiocarbon in the environment as a chronometer

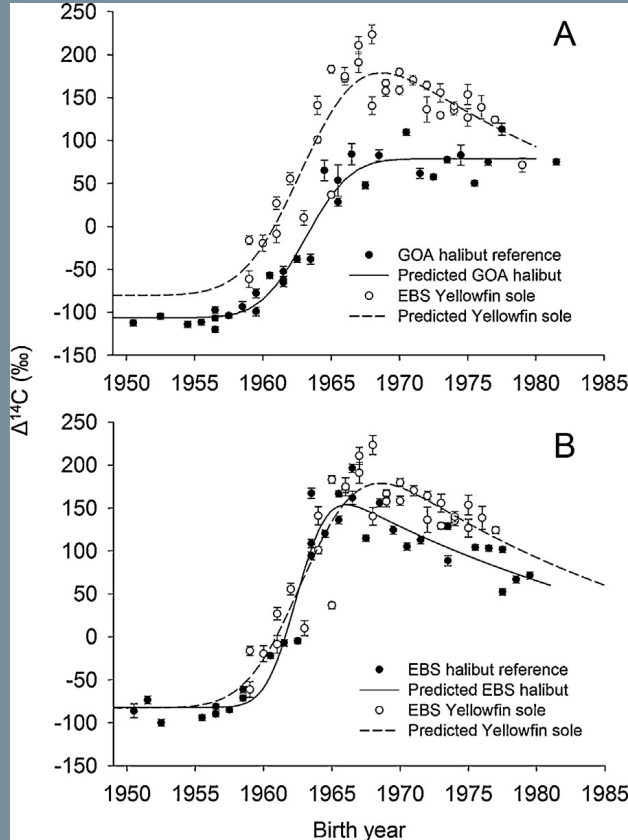
13C: stable isotope signal of habitat or region

15N: stable isotope signal of trophic position

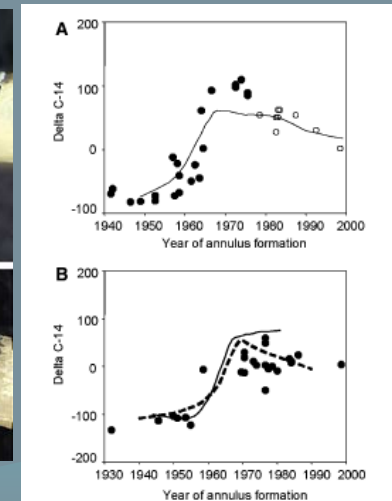
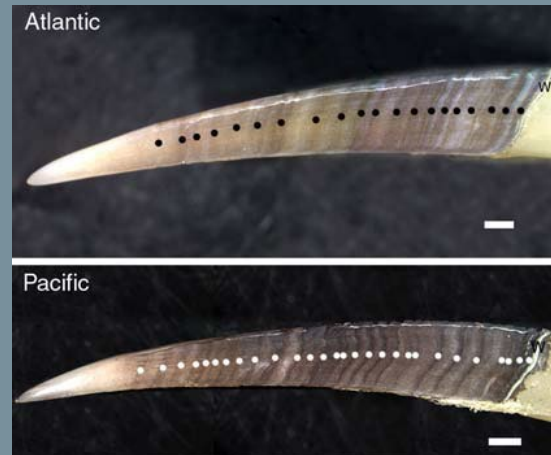
~~CSIA-AA: compound specific isotope analysis of amino acids with high resolution~~



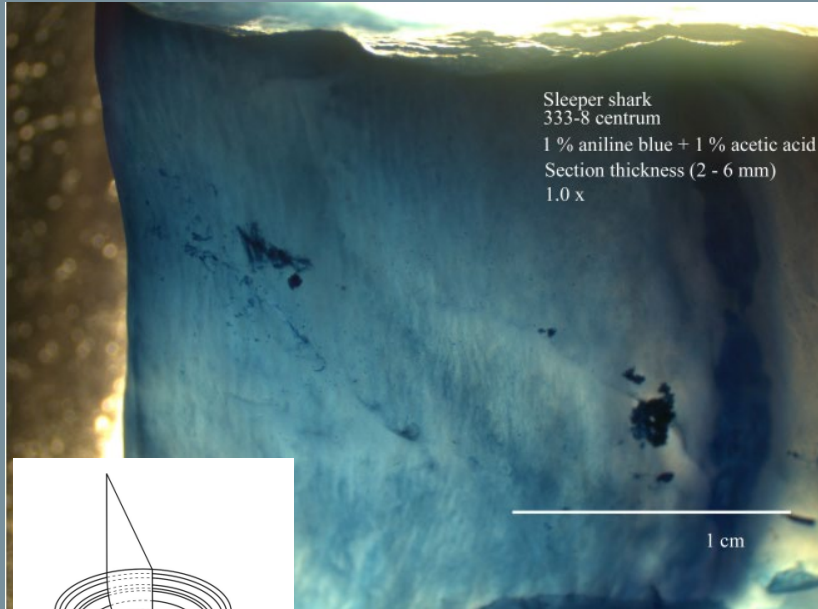
14C age validation of teleosts and sharks



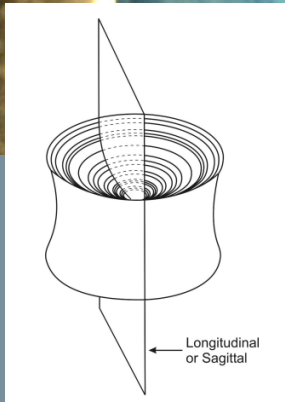
- Often used in teleosts and elasmobranchs to validate ageing methods
- Applied to species which HAVE ageing methods



14C for Sleeper Sharks



- Attempts at ageing Somniosids (aka sleeper sharks) with traditional methods have all failed



^{14}C for Sleeper Sharks



- Attempts at ageing Somniosids (aka sleeper sharks) with traditional methods have all failed
- Using bomb radiocarbon methods to establish longevity

Nielsen et al. 2016

14C for Sleeper Sharks

Are pet clinical trials ready
for prime time? p. 638

The sense (and antisense)
of ALS therapy pp. 647 & 708

Keeping fam
their homes

Science

\$15
12 AUGUST
sciencemag.org

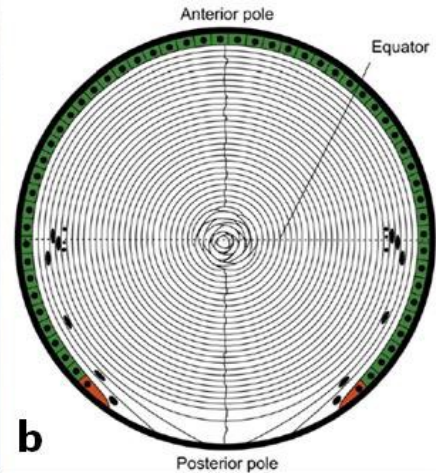


LONG IN THE TOOTH

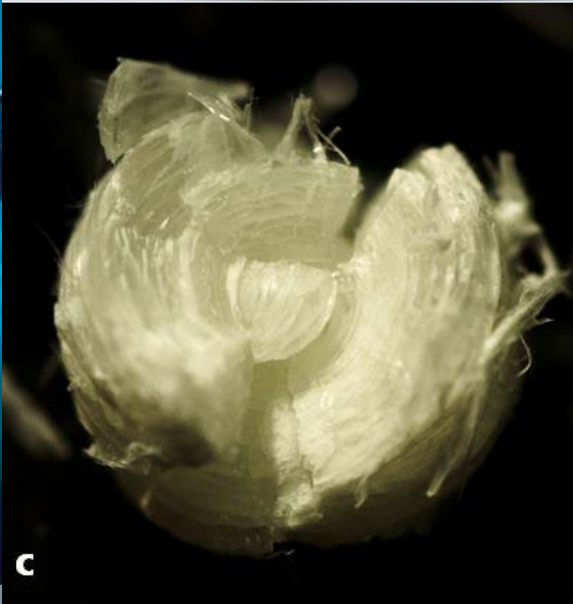
400-year-old sharks in
Arctic waters p. 702



a



b



c



d

14C for Sleeper Sharks

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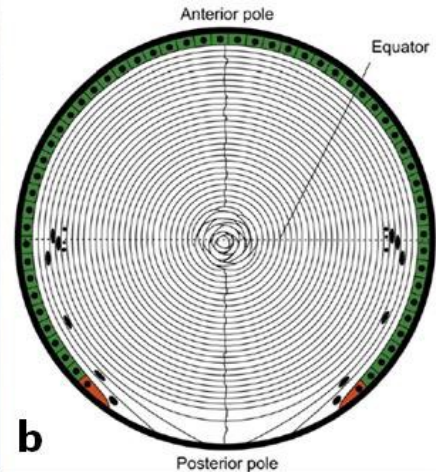


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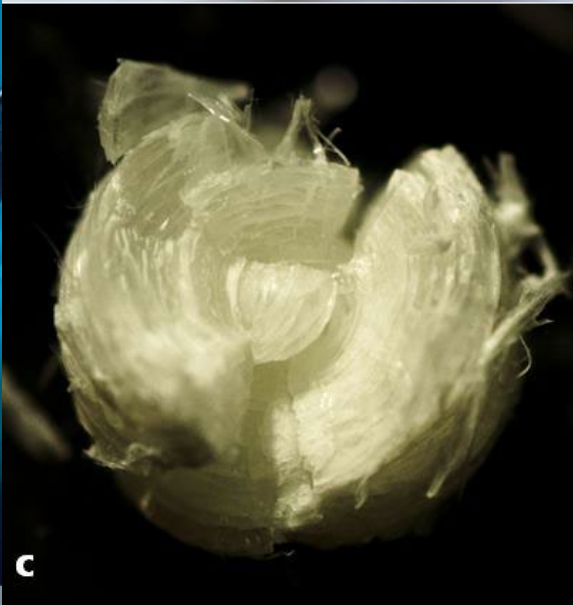
400-year-old sharks in
Arctic waters p. 702



a



b



c



d

NPRB 2301: Ageing of Pacific Sleeper Shark

Launching from the lessons learned from Nielsen et al. 2016

- Large sample size
- Lifetime profiles
- 40 - 465 cm TL
- Each eye layer retained for chemical analysis
- 14C, 13C, 15N and CSIAA-A
- Spiny dogfish with pups for comparison

Peeling eye layers

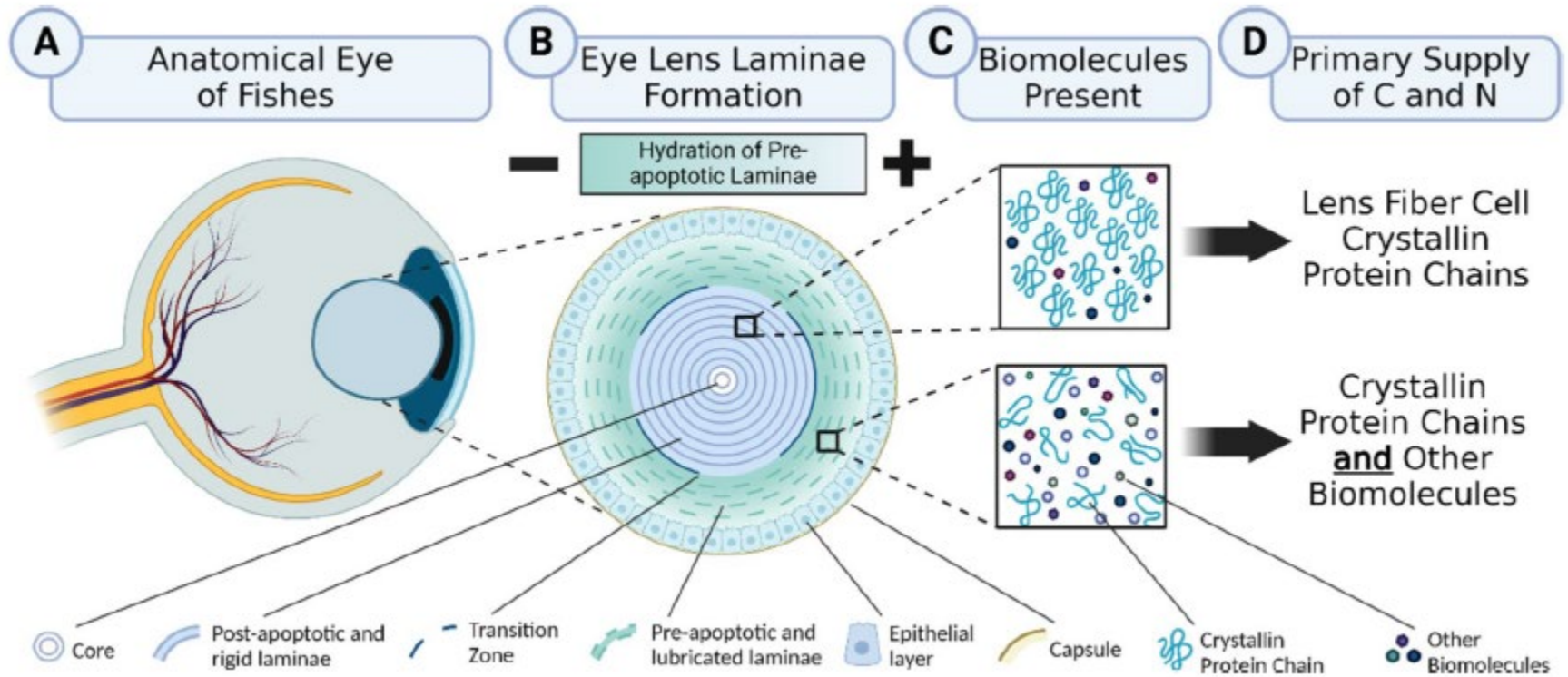
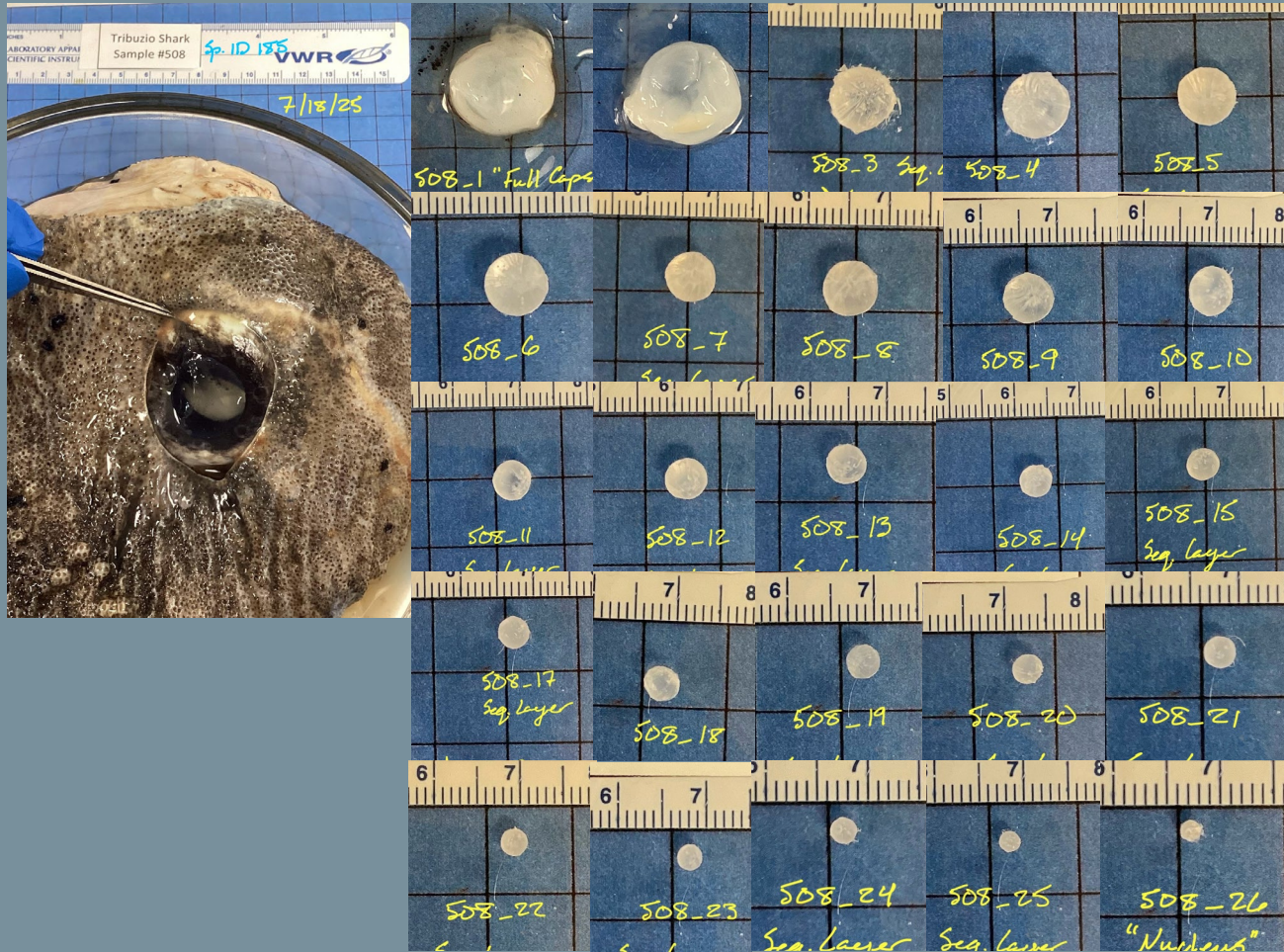


Fig. 1 Kuntz et al. 2025

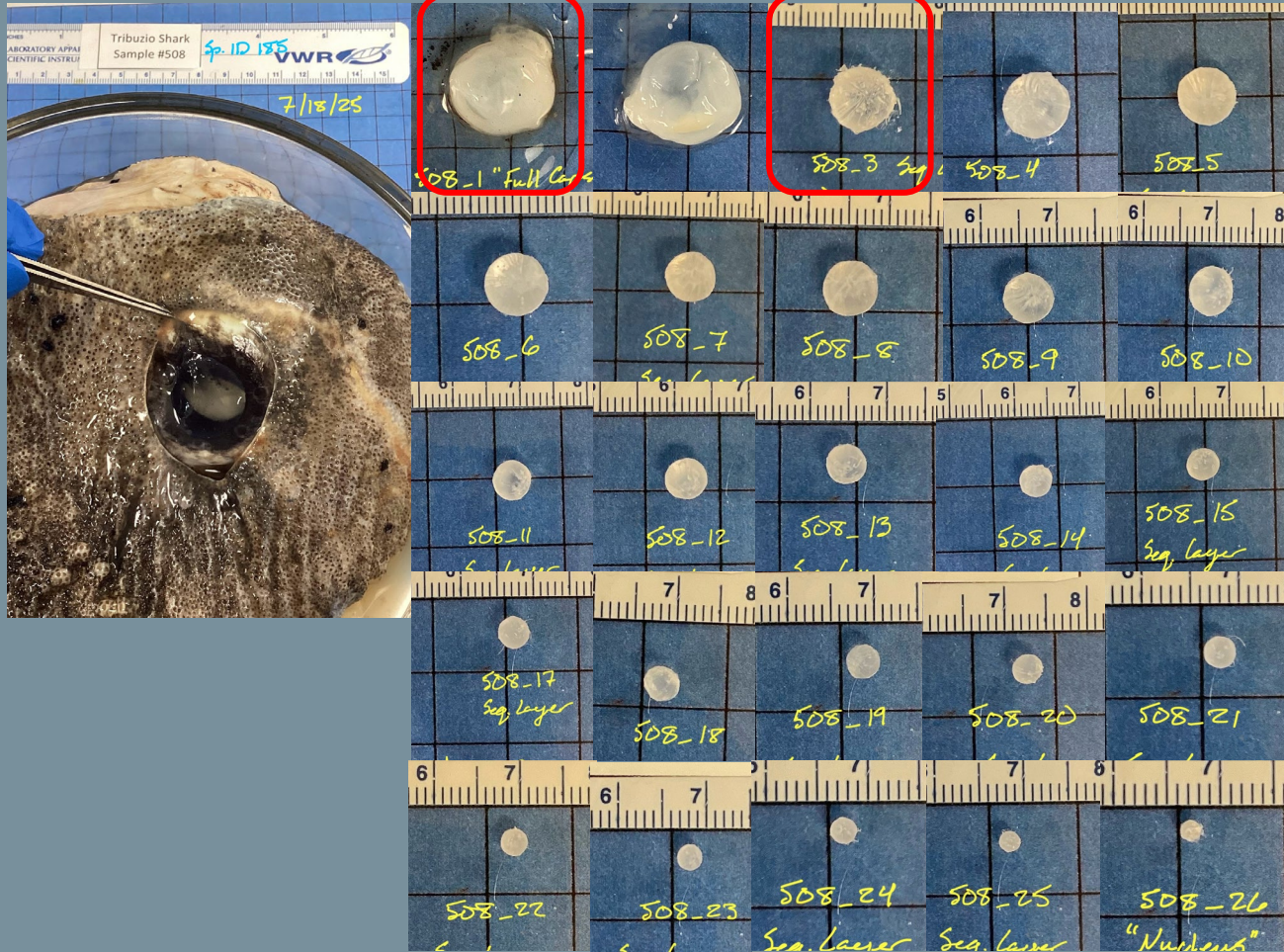
Peeling eye layers



465cm TL Female



Peeling eye layers



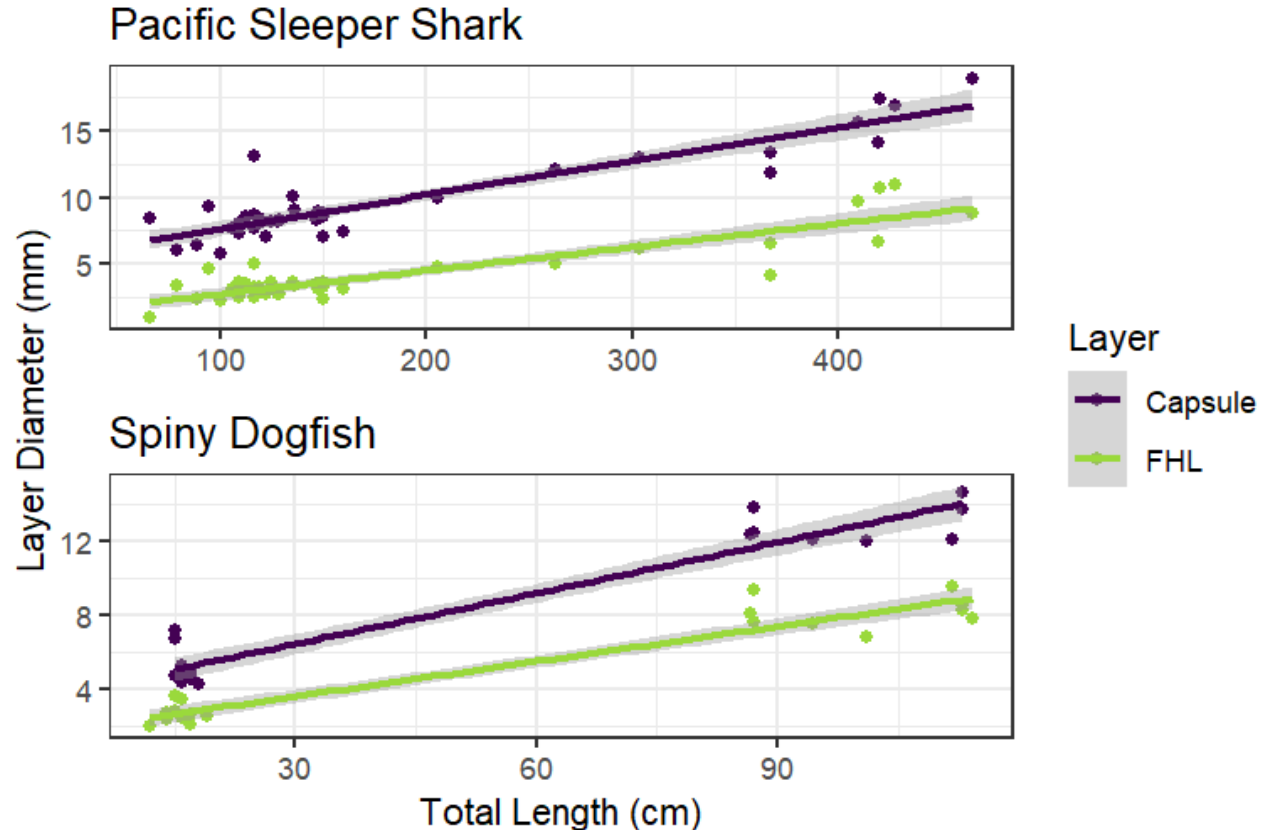
465cm TL Female



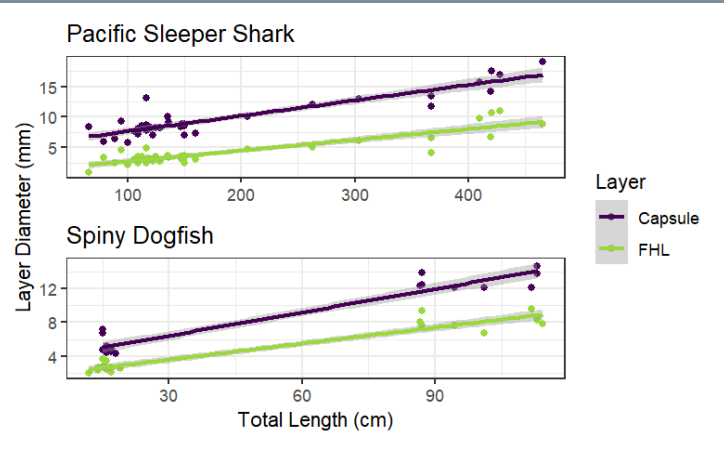
NPRB 2301: Preliminary Results

Linear relationships

Different slopes



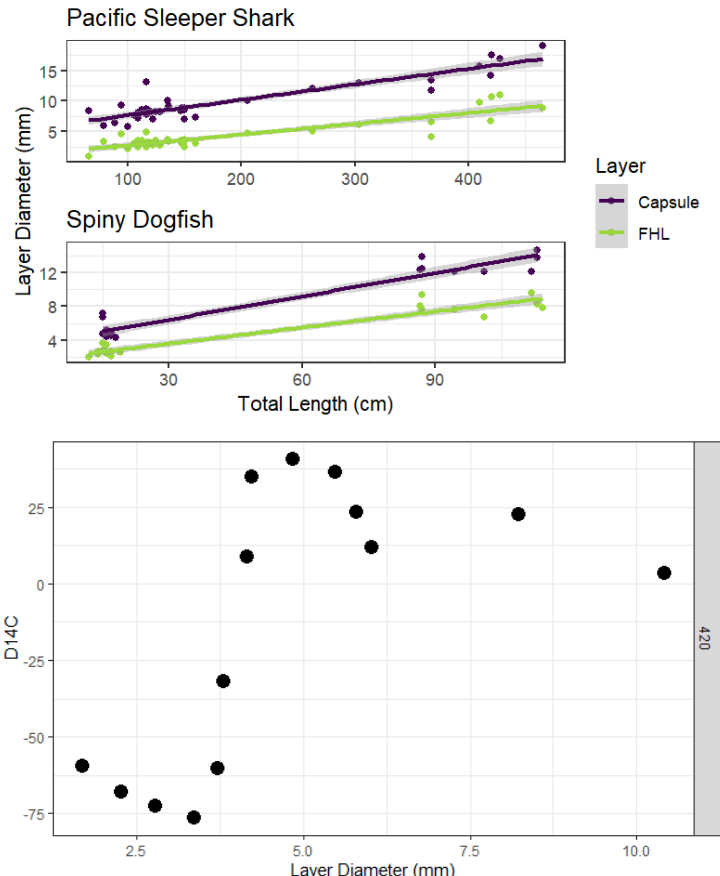
NPRB 2301: Preliminary Results



Linear relationship between eye lens diameter and total length

Diameter can be used to estimate length at time of layer formation

NPRB 2301: Preliminary Results



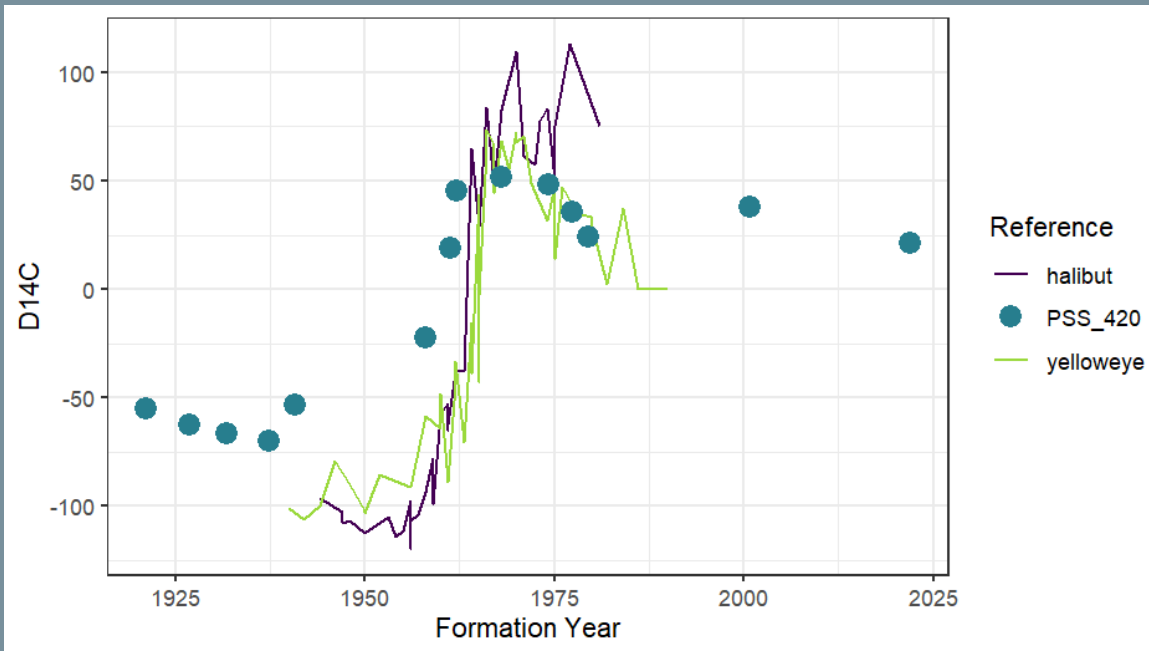
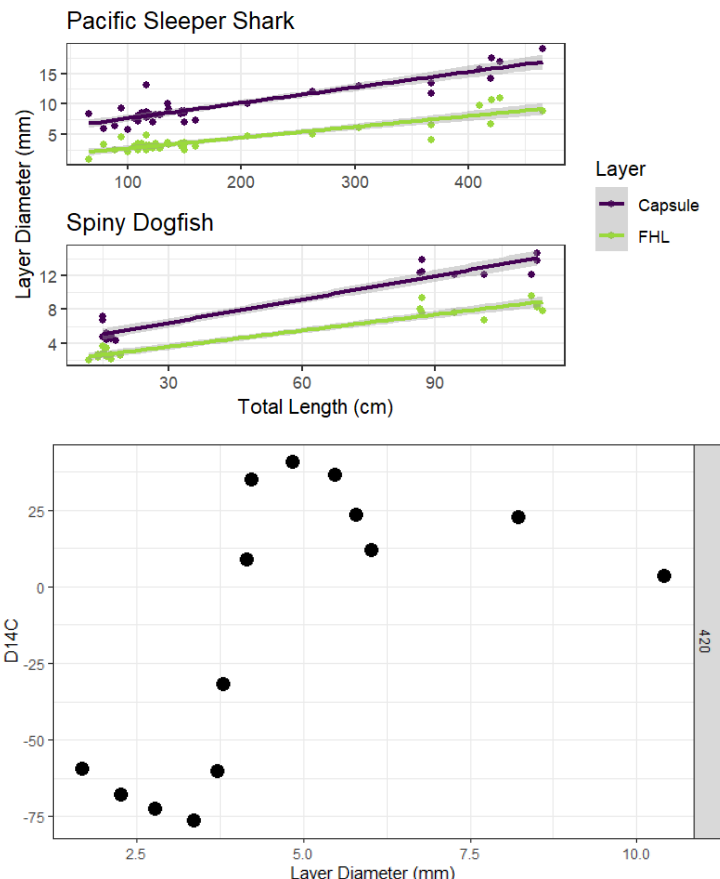
Linear relationship between eye lens diameter and total length

Diameter can be used to estimate length at time of layer formation

Proxy for time

420 cm TL, GOA, 2022

NPRB 2301: Preliminary Results

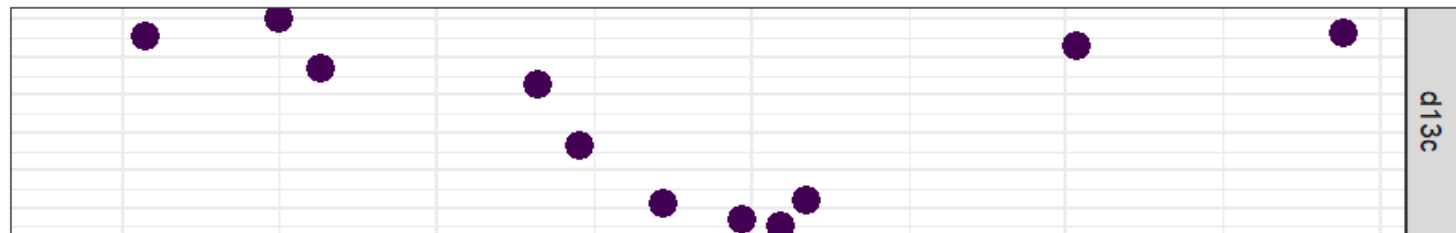


420 cm TL, GOA, 2022

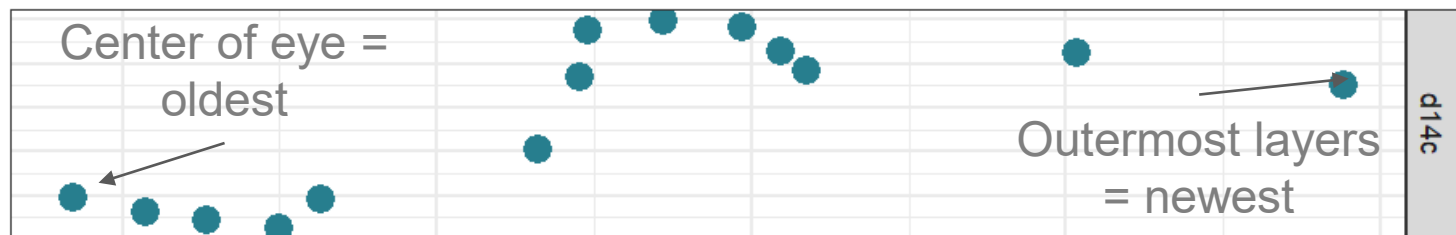


NPRB 2301: Preliminary Results

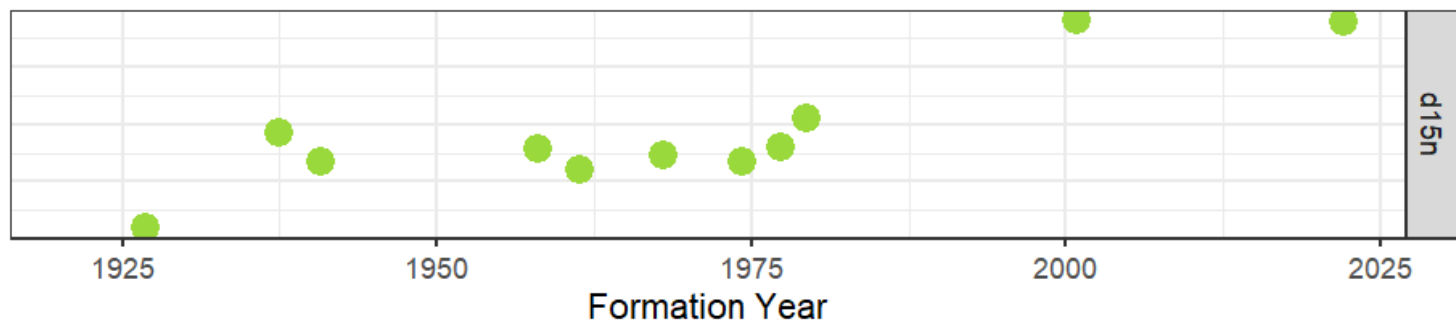
13C indicator
habitat or region



14C indicator
of age



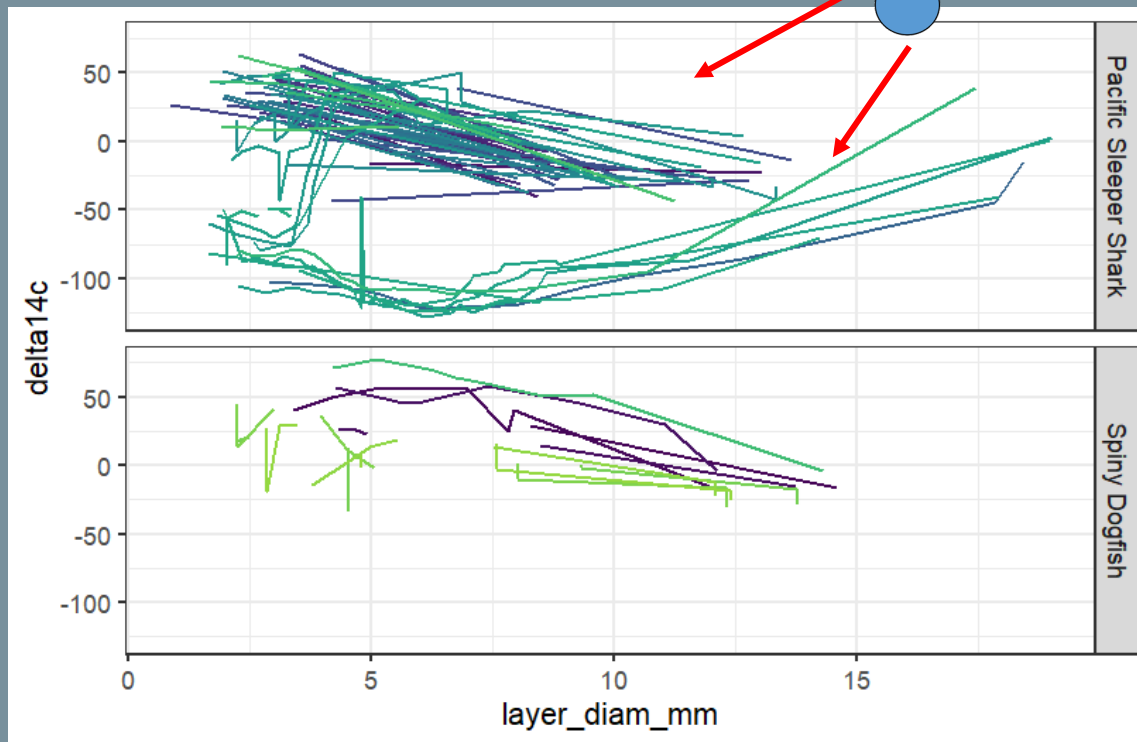
15N indicator
trophic position





NPRB 2301: Preliminary Results

Some sample sleuthing
needed





NPRB 2301: Next Steps

Some sample sleuthing
needed

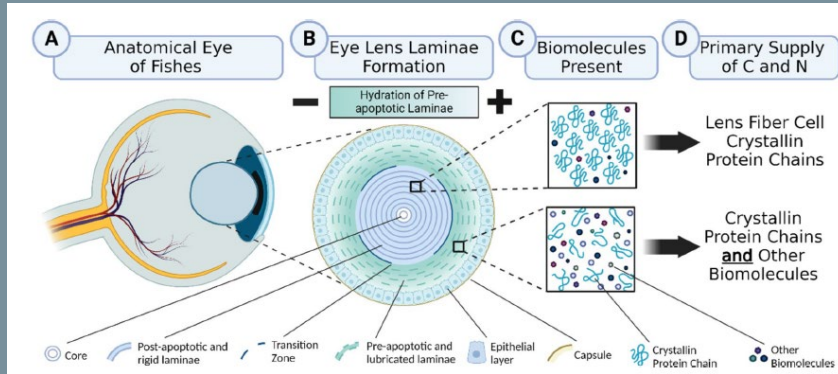
Sequential analytics needed
first

- Metadata
- Total length estimates
- Protein fractionation

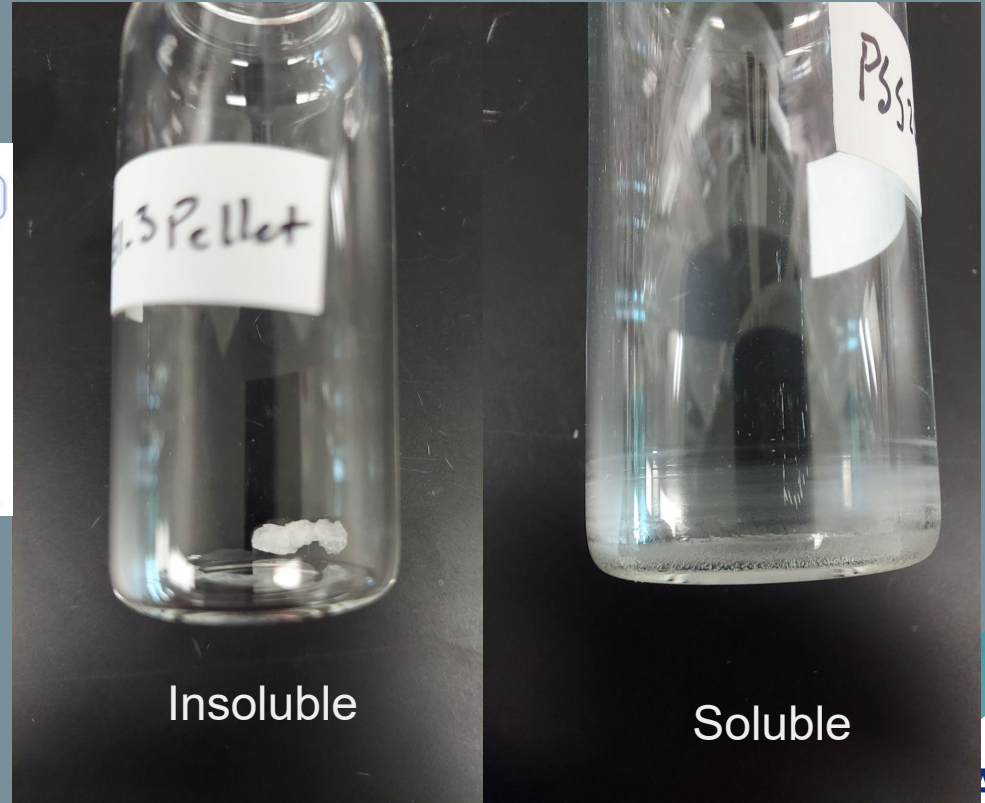


NPRB 2301: Next Steps

Each layer composed of soluble and insoluble proteins



Modern carbon can infiltrate the soluble protein in the layers, potentially at different rates



NPRB 2301: Next Steps

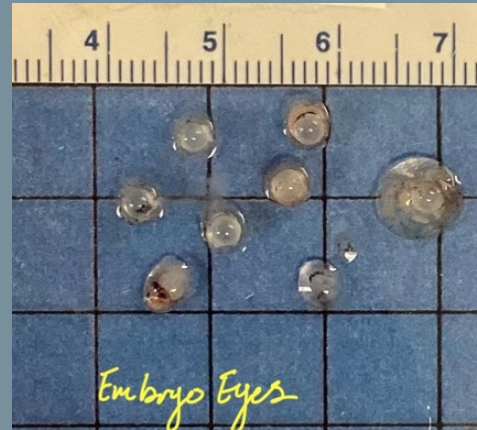
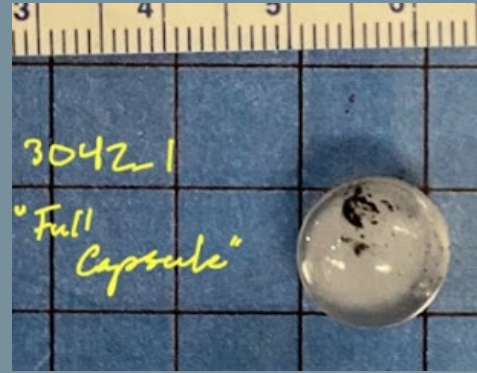
(Obj 2) Finish length estimation and protein fractionation

(Obj 1) Assign year ranges to eye layers

(Obj 1 & 2) Compare/contrast with spiny dogfish and maternal/offspring pairs

(Obj 1 & 2) Interpret 14C, SIA and CSIA-AA results (that's all)

(Obj 3) Feedback to management



NPRB 2301: Next Steps

Advance the stock assessment
by informing:



Longevity



Growth rate



Natural mortality



Exploration of expanded
data-limited assessment
methods



Demographic simulations



Risk table levels of concern



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Thanks!!!

North Pacific Research Board for funding and next level support to solve logistical problems

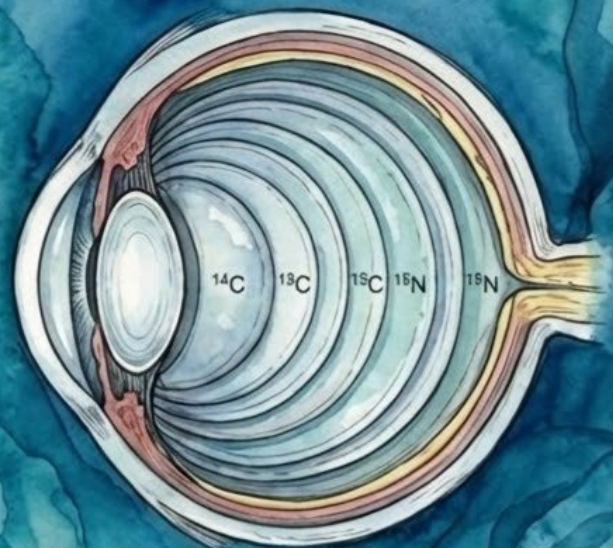
ABL RECA program for lab space and their patience as we figured out our procedures and ABL supervisors for fighting the good fight

All of the at-sea observers, The North Pacific Groundfish Observer Program, and At-Sea Hake Observer Program for supporting the collections

Survey science and vessel crew for taking the time to sample

Fishing vessels who provided samples

Sample processing at UC Santa Cruz Stable Isotope Laboratory, Stanford Stable Isotope Biogeochemistry Laboratory and WHOI NOSAMS



Gemini is trying its hardest, but just can NOT make a sleeper shark

For further questions, contact cindy.tribuzio@noaa.gov

